

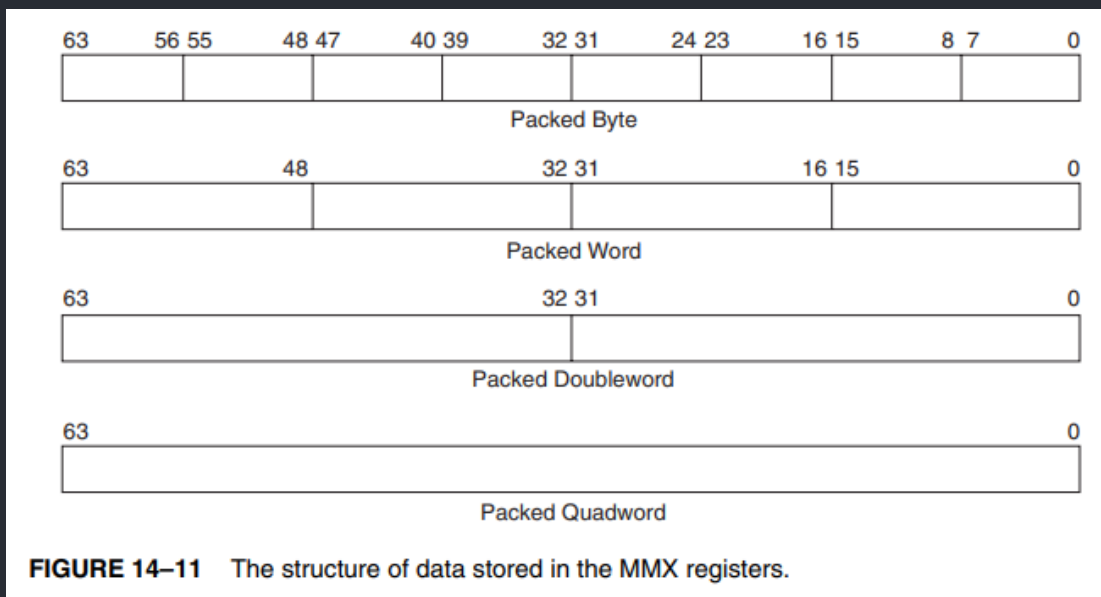
سوالات تحلیلی

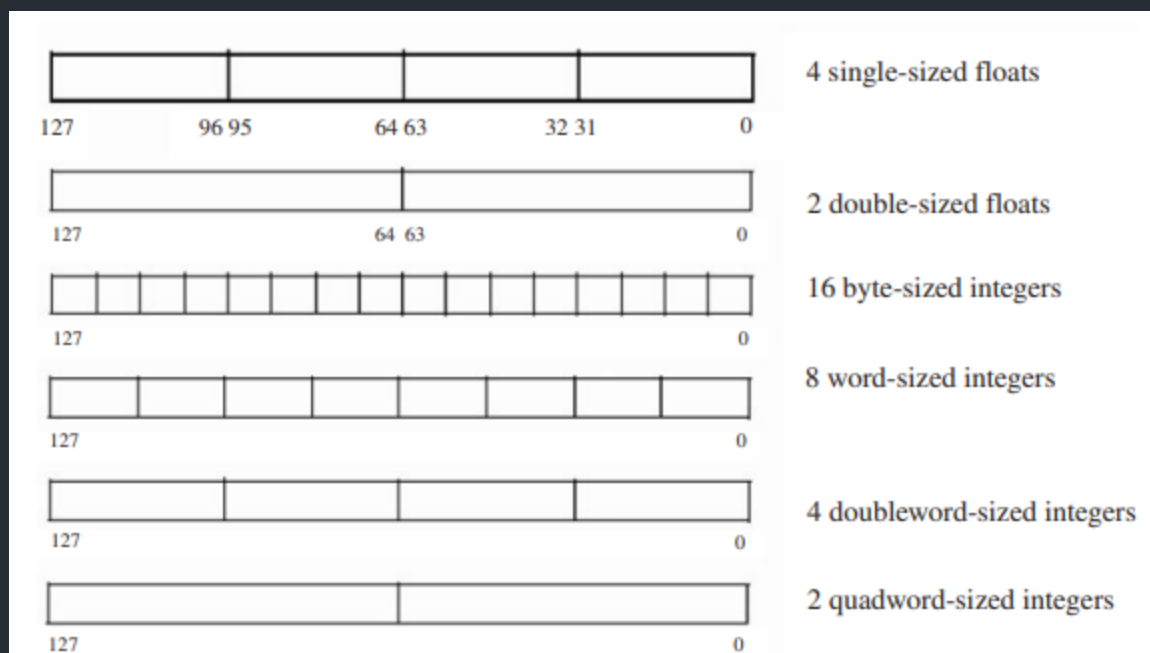
سوال ۱: در سیر تکاملی پردازنده‌ها چه راهکارهایی برای غلبه بر چالش گلوگاه بودن پهنای باند حافظه پیشنهاد شده است؟

- Memory Hierarchy and Caches
- Memory Controllers
- Memory Interleaving
- Memory Channeling
- SIMD ISA
- Memory Compression

سوال ۲: با مراجعه به مرجع Brey شباهت‌ها و تفاوت‌ها بین دو فناوری MMX و SSE را شرح دهید.

- Similarities:
 - SIMD Architecture
 - new Register Set: MMX introduces eight 64-bit MMX registers, while SSE initially introduced eight 128-bit XMM registers and later expanded to 16 XMM registers with SSE2.
- Differences:
 - Data Width: MMX operates on 64-bit packed integer data, whereas SSE operates on 128-bit packed integer or floating-point data.





- Instruction Set
- SSE Floating-Point Support

گزارش دستور کار:

کد:

```

Lab12.cpp  [X]
Lab12 (Global Scope)
1  // Lab12.cpp : This file contains the 'main' function. Program execution begins and ends there.
2  //
3
4  #include <conio.h>
5  #include <cstdlib>
6  #include <chrono>
7  using namespace std::chrono;
8
9  void print_array(float* arr, int size, bool is_float) {
10     _cprintf_s("[");
11     for (int i = 0; i < size; i++) {
12         if (is_float) {
13             _cprintf_s("%.2f, ", arr[i]);
14         }
15         else {
16             _cprintf_s("%.2E, ", arr[i]);
17         }
18     }
19     _cprintf_s("]\n");
20 }
21

```

```

Lab12.cpp  [X]
Lab12 (Global Scope)
22 int main()
23 {
24     float a, b;
25     float y[50];
26     float zi1[50], zi2[50], zo1[50], zo2[50];
27
28     _cprintf_s("Enter a: ");
29     _cscanf_s("%f", &a);
30     _cprintf_s("\n");
31     _cprintf_s("Enter b: ");
32     _cscanf_s("%f", &b);
33     _cprintf_s("\n");
34
35     for (int i = 0; i < 50; i++) {
36         y[i] = 0 + (0.2 * i);
37     }
38
39     _cprintf_s("y: ");
40     print_array(y, 50, true);
41
42     for (int i = 0; i < 50; i++) {
43         zi1[i] = y[i]*y[i] - a;
44         zi2[i] = y[i]*y[i] + 2 * b;
45     }
46
47     _cprintf_s("\nzi1: ");
48     print_array(zi1, 50, true);
49     _cprintf_s("\nzi2: ");
50     print_array(zi2, 50, true);

```

```

Lab12.cpp  [X]
Lab12 (Global Scope)
51
52     auto start = high_resolution_clock::now();
53
54     for (int i = 0; i < 50; i++) {
55         zol[i] = 1 / (2 * zi1[i] * zi2[i]);
56     }
57
58     auto stop = high_resolution_clock::now();
59     auto duration = duration_cast<microseconds>(stop - start);
60
61     _cprintf_s("\nTime taken by function: %f microseconds", duration.count());
62     _cprintf_s("\nzol: ");
63     print_array(zol, 50, false);
64     _cprintf_s("\n");
65

```

```

Lab12.cpp  [X]
Lab12 (Global Scope)
66
67     start = high_resolution_clock::now();
68
69     __asm {
70         mov ecx, 0
71     LOOP1:
72         movups xmm0, oword ptr zi1[4 * ecx]
73         movups xmm1, oword ptr zi2[4 * ecx]
74
75         movups xmm2, xmm0
76         mulps xmm2, xmm1
77         addps xmm2, xmm2
78         rcpps xmm3, xmm2
79         movups oword ptr zo2[4 * ecx], xmm3
80
81         add ecx, 4
82         cmp ecx, 50
83         jl LOOP1
84     }
85
86     stop = high_resolution_clock::now();
87     duration = duration_cast<microseconds>(stop - start);
88
89     _cprintf_s("Time taken by function: %f microseconds", duration.count());
90     _cprintf_s("\nzo2: ");
91     print_array(zo2, 50, false);
92     _cprintf_s("\n");
93

```

تست:

```

C:\Users\ROG\Documents\Erfun\University\Semester 5th\Microprocessors and Interface Circuits\HW\Micro-Archive\Lab12\Visual Stu...
Enter a: 2.2
Enter b: 2.5
y: [0.00, 0.20, 0.40, 0.60, 0.80, 1.00, 1.20, 1.40, 1.60, 1.80, 2.00, 2.20, 2.40, 2.60, 2.80, 3.00, 3.20, 3.40, 3.60, 3.80, 4.00, 4.20, 4.40, 4.60, 4.80, 5.00, 5.20, 5.40, 5.60, 5.80, 6.00, 6.20, 6.40, 6.60, 6.80, 7.00, 7.20, 7.40, 7.60, 7.80, 8.00, 8.20, 8.40, 8.60, 8.80, 9.00, 9.20, 9.40, 9.60, 9.80, ]

zi1: [-2.20, -2.16, -2.04, -1.84, -1.56, -1.20, -0.76, -0.24, 0.36, 1.04, 1.80, 2.64, 3.56, 4.56, 5.64, 6.80, 8.04, 9.36, 10.76, 12.24, 13.80, 15.44, 17.16, 18.96, 20.84, 22.80, 24.84, 26.96, 29.16, 31.44, 33.80, 36.24, 38.76, 41.36, 44.04, 46.80, 49.64, 52.56, 55.56, 58.64, 61.80, 65.04, 68.36, 71.76, 75.24, 78.80, 82.44, 86.16, 89.96, 93.84, ]

zi2: [5.00, 5.04, 5.16, 5.36, 5.64, 6.00, 6.44, 6.96, 7.56, 8.24, 9.00, 9.84, 10.76, 11.76, 12.84, 14.00, 15.24, 16.56, 17.96, 19.44, 21.00, 22.64, 24.36, 26.16, 28.04, 30.00, 32.04, 34.16, 36.36, 38.64, 41.00, 43.44, 45.96, 48.56, 51.24, 54.00, 56.84, 59.76, 62.76, 65.84, 69.00, 72.24, 75.56, 78.96, 82.44, 86.00, 89.64, 93.36, 97.16, 101.04, ]

Time taken by function: 0.000000 microseconds
zo1: [-4.55E-02, -4.59E-02, -4.75E-02, -5.07E-02, -5.68E-02, -6.94E-02, -1.02E-01, -2.99E-01, 1.84E-01, 5.83E-02, 3.09E-02, 1.92E-02, 1.31E-02, 9.32E-03, 6.90E-03, 5.25E-03, 4.08E-03, 3.23E-03, 2.59E-03, 2.10E-03, 1.73E-03, 1.43E-03, 1.20E-03, 1.01E-03, 8.56E-04, 7.31E-04, 6.28E-04, 5.43E-04, 4.72E-04, 4.12E-04, 3.61E-04, 3.18E-04, 2.81E-04, 2.49E-04, 2.22E-04, 1.98E-04, 1.77E-04, 1.59E-04, 1.43E-04, 1.30E-04, 1.17E-04, 1.06E-04, 9.68E-05, 8.82E-05, 8.06E-05, 7.38E-05, 6.77E-05, 6.22E-05, 5.72E-05, 5.27E-05, ]

Time taken by function: 0.000000 microseconds
zo2: [-4.54E-02, -4.59E-02, -4.75E-02, -5.07E-02, -5.68E-02, -6.95E-02, -1.02E-01, -2.99E-01, 1.84E-01, 5.83E-02, 3.09E-02, 1.92E-02, 1.31E-02, 9.32E-03, 6.91E-03, 5.25E-03, 4.08E-03, 3.23E-03, 2.59E-03, 2.10E-03, 1.73E-03, 1.43E-03, 1.20E-03, 1.01E-03, 8.56E-04, 7.31E-04, 6.28E-04, 5.43E-04, 4.72E-04, 4.12E-04, 3.61E-04, 3.18E-04, 2.81E-04, 2.49E-04, 2.22E-04, 1.98E-04, 1.77E-04, 1.59E-04, 1.43E-04, 1.29E-04, 1.17E-04, 1.06E-04, 9.68E-05, 8.82E-05, 8.06E-05, 7.38E-05, 6.77E-05, 6.22E-05, 5.72E-05, 5.27E-05, ]

```

